

Therapeutic efficacy of once-daily oral administration of a Kunitz-type protease inhibitor, bikunin, in a mouse model and in human cancer.

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BACKGROUND: Bikunin, a Kunitz-type protease inhibitor, specifically inhibits tumor invasion and metastasis. METHODS: The authors initially evaluated the therapeutic efficacy of once-daily oral administration of different doses of bikunin against human ovarian carcinoma HRA cells growing in the peritonea of nude mice. For the in vivo studies, female 7-week-old nude mice were randomized to 1 of 4 groups: bikunin-treated groups (n = 9 in each group) received 3, 10, or 30 microg/g body weight per day bikunin for 7 days via gastrointestinal gavage, and a control group (n = 9) received the vehicle solution (phosphate-buffered saline) via gastrointestinal gavage. On Day 9, the abdominal cavity was examined by two observers who were blinded to treatment. RESULTS: After oral administration, intact bikunin was detectable in mouse serum specimens at 3 and 6 hours. This was followed by a decline at 12 hours. The mice given bikunin at the highest dose level had a 40% decrease in tumor load. The highest uptake in the tumor was obtained with [¹²⁵I]bikunin 12 hours postadministration. No effect on either food intake or body weight was observed in the treated versus sham groups. The current study was the first to report the potent activity of once-daily oral administration of bikunin against ovarian carcinoma. Next, the authors performed a Phase I trial to determine the maximum-tolerated dose (MTD) and safety of a once-daily oral administration schedule. The indication was locally advanced uterine cervical carcinoma after definitive treatment. An escalating dose (3, 10, and 30 mg/kg per day) of bikunin was administered orally to nine patients for 7 days. There were no dose-limiting toxicities and the MTD of the bikunin schedule was not defined. The authors also obtained preliminary data on its effect on urokinase-type plasminogen activator expression at the highest dose level. CONCLUSIONS: Once-daily oral administration of bikunin was found to be safe in humans and exhibited signs of biologic activity. Copyright 2004 American Cancer Society.